



AUBURN UNIVERSITY

WILD 7450 Section: 001

Tools & Challenges for Science

Fall 2026

Faculty Information

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Course Information

Credit Hours: 2

Contact Hours: 2

Catalog Description: Contemporary tools used and challenges encountered by modern scientists and how these factors have altered the process of science.

Specific topics will be crowdsourced and tailored to students' interests, but examples include: diversity in science, reproducibility crisis, open science practices, version control, online teaching, artificial intelligence, work expectations, ethics of internships and volunteer positions, citizen science, modern computing power, cloud computing, self promotion, and mental health in academia

Modality: Lecture

Meeting Information

Meeting Times: 1:00 PM-2:50 PM

Meeting Days: FRI

Meeting Location: Forestry & Wildlife Sci Bldg 1219

Course Description

This is a 2-hour discussion-based class focused on the contemporary tools and challenges encountered by modern scientists and how these factors have altered the process of science. Specific topics will be crowdsourced and tailored to students' interests, although we have provided several examples below. Students will be able to add topics to the sections below, based on their interests, and then we will select several topics per section for discussion during the semester.

Course Objectives

The overall goals of this course will be to: (1) help students develop skills for navigating a modern science career; and (2) spark dialogue about how the practice of science is being altered by contemporary forces, for better or worse.

Student Learning Outcomes

- Students will demonstrate critical thinking skills by applying knowledge from readings and personal experiences to classroom discussions.
- Students will demonstrate understanding of the tools available to them for science communication through discussion and ultimately using these tools to promote their own research.
- When presented with options for potential publishing outlets, students will be able to assess those options critically, responsibly, and ethically.
- Through discussions, students will demonstrate understanding of the philosophical nuances associated with different types of data and analytical approaches.
- Students will be able to identify tools available to them to improve the reproducibility of their research and will demonstrate knowledge of the advantages and limitations of these tools via in-class discussions and out-of-class applications.

Required Materials

Title: None

Assessments, Exams, & Evaluations

Assignments:

Classroom participation – this is a discussion course, and you are expected to:

- Prepare for each class by reading/viewing the assigned materials before class;
- Attend class; and
- Participate in in-class discussions.

Weekly written assignment – after every class period, you will be given an assignment in which you will provide written feedback on the topic. We will provide a prompt for you (e.g., write a letter to a politician summarizing your position on the topic).

- Weekly assignments will be due by the start of the following week's class period.
- These assignments will be maintained in an R Quarto document that is stored and shared on GitHub.

Presentation – Once during the semester, each student will be responsible for leading the class discussion.

The presenter will identify the primary article on the topic s/he would like the class to read.

- You MUST have this reading approved by the instructors before assigning it.
- You MUST have the reading available to students by the time class starts the week before your presentation
- Therefore, you MUST get your assignment approved more than 1-week ahead of time.

The presenter will provide an ~ 20-minute summary presentation on the topic via PowerPoint and facilitate a classroom discussion

The presenter will generate 5 discussion questions and the class will be separated to discuss these in small groups

We will then return to a larger group for the final hour of class for a group discussion

Finally, the presenter will develop a thought-provoking writing prompt for the class.

Grading Standards & Attendance Expectations

Grading and Evaluation Procedures:

- 90-100% is an A, 80-89% is a B, 70-79% is a C, 60-69% is a D, and grades < 60% will earn an F.
- Preparation for and participation in conversation topics led by other students will comprise 34% of the student's grade.
- Performance on weekly written assignments will comprise 33% of the grade.
- The final 33% of the grade will be based on the student's presentation and facilitation of in-class discussion.

Attendance Expectations:

Weekly attendance is critical given that participation in discussions comprise 34% of the student's grade.

- Students are allowed 1 excuse-free absence from class during the semester
- Beyond that, students will be granted excused absences from class with appropriate documentation for the following reasons: illness of the student or serious illness of a member of the student's immediate family, death of a member of the student's immediate family, trips for student organizations sponsored by an academic unit, trips for University classes, trips for participation in intercollegiate athletic events, subpoena for a court appearance and religious holidays.
- Students who wish to have an excused absence from this class for any other reason must contact the instructors in advance of the absence to request permission. The instructors will weigh the merits of the request and render a decision. When feasible, the student must notify the instructors prior to the occurrence of any excused absences, but in no case shall such notification occur more than one week after the absence. Appropriate documentation for all excused absences is required.
- Because learning via discussion is the crux of this class, any absence

(beyond the first) will require preparation of an additional 1,000-word written assignment that summarizes additional literature on the topic.

Academic Integrity

Auburn University upholds the highest standards of honesty and integrity in all academic pursuits. Students are expected to conduct themselves in accordance with these principles at all times. Please review Auburn University's Academic Integrity [Policy](#) to familiarize yourself with actions that constitute violations of this policy. Any alleged violations will be reported to the Office of the Provost for review and appropriate action. The Academic Integrity Policy is applicable to all students taking classes at Auburn University with the exception of students that are enrolled in the Doctor of Veterinary Medicine or Doctor of Pharmacy programs.

Generative Artificial Intelligence Tools

AI Policy: Not Permitted in this Course for Assignments

In this course, it is expected that all submitted work is produced by the students themselves, whether individually or collaboratively. Students must not seek the assistance of Generative AI Tools like Microsoft Copilot, ChatGPT, Claude, or Gemini for graded assignments. Use of a Generative AI Tool to complete an assignment constitutes academic dishonesty. Students may use Generative AI tools as a study tool, but be forewarned that AI tools are not trustworthy.

ADA Compliance

Students who need accommodations should submit their approved accommodations through the AIM Student Portal on AU-Access and communicate directly with faculty about accommodations for their course. It is important for the student to complete these steps as soon as possible; accommodations are not retroactive. Students who have not established

accommodations through the Office of Accessibility but need accommodations should contact the Office of Accessibility at ACCESSIBILITY@auburn.edu or (334) 844-2096 (V/TT). The Office of Accessibility is located in Haley Center 1228.

Equal Opportunity Compliance

If you have experienced discrimination, harassment, sexual misconduct, relationship violence, stalking, or another situation affecting your ability to participate fully in the Auburn University community, the Office of Equal Opportunity Compliance (EOC) is available to help. EOC also coordinates accommodations related to religious observances, pregnancy, and related medical conditions. Information about EOC policies, resources, and religious and pregnancy-related accommodations, including request forms and a map of campus lactation spaces, is available at aub.ie/eoc.

Auburn's non-discrimination and anti-harassment statement is available at this [website](#). Most Auburn University faculty and staff are required to share information about possible discrimination, harassment, retaliation, or sexual misconduct with EOC so that support, resources, and information about available options can be offered to those who may be impacted. Receiving a report does not automatically result in a formal complaint or investigation. Students seeking confidential support may contact Safe Harbor (334-844-7233) or Student Counseling & Psychological Services (334-844-5123). To learn more about EOC services, reporting options, supportive measures, and accommodations, visit aub.ie/eoc, email eoc@auburn.edu, or call 334-844-4794.

Student Conduct

Student conduct promotes a safe, respectful, and accountable campus environment by upholding the [Code of Student Conduct](#) and addressing behaviors that affect the Auburn community. Students may contact the office with non-academic questions or concerns, report potential policy violations, or submit a written student complaint by calling 334-844-1305 or visiting auburn.edu/

[studentconduct](#).

Auburn Cares

Do you need support or are you concerned about another person? Auburn Cares is here for you as you navigate academic, financial, mental health, or relationship issues. Auburn Cares is a good first point of contact for students experiencing a difficult situation. Contact Auburn Cares at 334-844-1305 or aub.ie/auburncares to request support today for yourself or another person.

Student Counseling & Psychological Services

Student Counseling & Psychological Services provides confidential, no-cost mental health counseling and psychiatric services to Auburn students. To schedule an appointment or speak with a counselor immediately 24/7/365, contact Student Counseling & Psychological Services at 334-844-5123.

Mental Health Emergencies

If you or someone else are in immediate danger, call 911. If you would like to speak with someone confidentially, Safe Harbor (334-844-7233) and Student Counseling & Psychological Services (334-844-5123) are both confidential resources. Safe Harbor provides support to students who have experienced sexual or relationship violence by connecting them with academic, medical, mental health, and safety resources. For additional information, visit auburn.edu/safeharbor.

Elasticity Statement

The instructor will make every effort to follow the guidelines of this syllabus as listed; however, the instructor reserves the right to amend this document as the need arises consistent with university policy. In such instances, the instructor will notify students in class and/or via email and will endeavor to provide reasonable time for students to adjust to any changes.

Changes to the syllabus may become necessary as the class progresses.

Updates to the course schedule will be provided to students in class and/or via email.

Use of Canvas or Other Technology Tools

Weekly assignments and other course notifications will be disseminated via Canvas. Students are responsible for checking Canvas to ensure they are up-to-date on course

information. Note that each student has control over her/his notifications via Canvas and can edit settings to alert them when an announcement is posted, an assignment is due, a grade is released, etc.